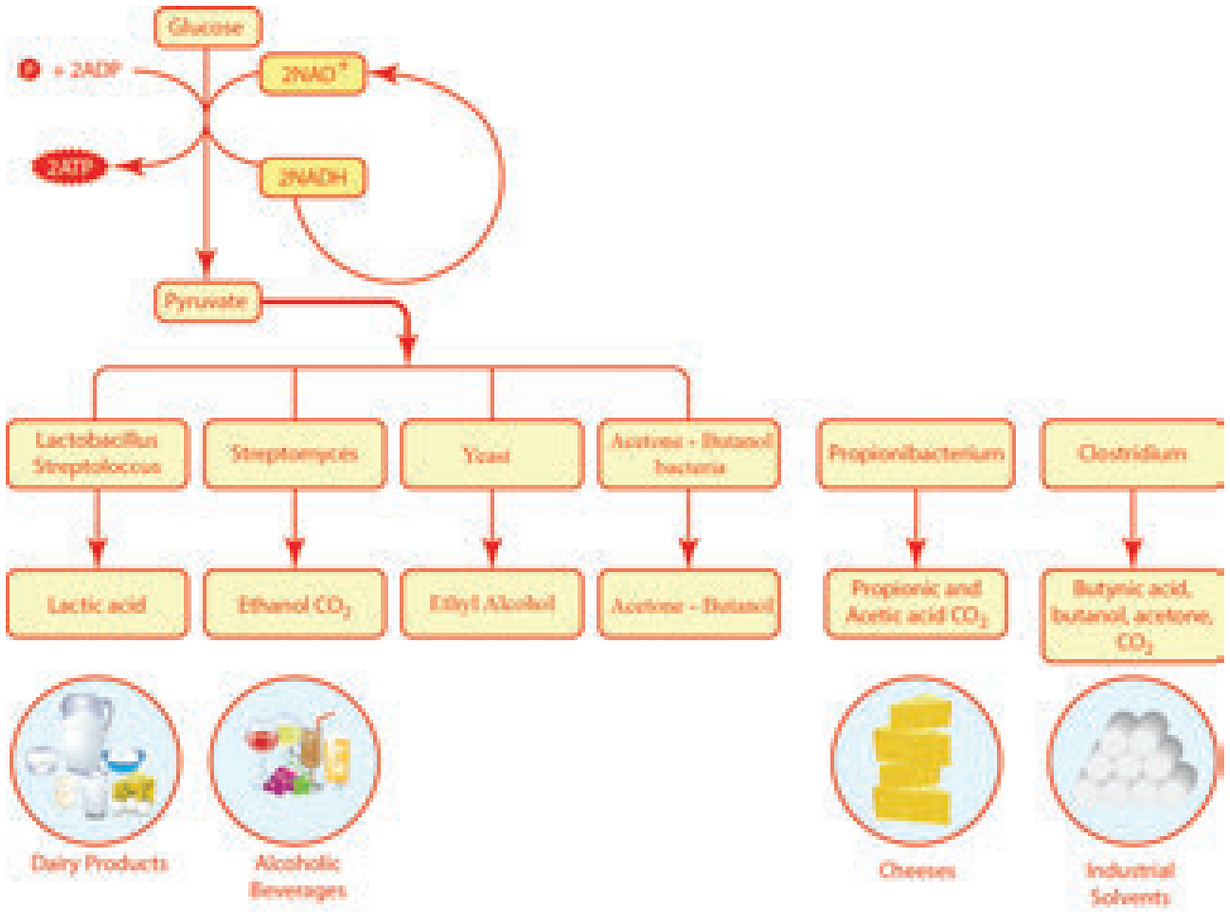


Figure 4.10: Varoius products of microbial fermentation

fermentation or heterolactic and often uses the pentose phosphate pathway.

$$G + ADP + (P) \longrightarrow \text{Lactic acid} + \text{ethanol} + CO_2 + ATP$$

HOTS

Why do cells need to ferment when they get 2ATPs from Glycolysis?

4.9.2 Alcohol Fermentation

Alcohol fermentation begins with the Glycolysis which yields two molecules of pyruvic acid and two molecules of ATPs. In the next step, the two molecules of pyruvic acid are converted to two molecules of acetaldehyde and two molecules of CO_2. The acetaldehydes are then reduced by NADH to form ethanol. The ethanol and CO_2 produced by the yeast Saccharomyces is used in alcoholic beverages and to raise bread dough respectively.

4.10 Enzymes

Life is an intricate meshwork involving a perfect coordination of a vast majority of chemical reactions. This is due to the presence of some catalysts synthesized inside the body of the organism. The term 'enzyme' was coined by Friedrich Wilhen Kuhne (1878) to designate these biological catalysts. The name 'enzyme' (en – in, zyme – yeast) literally means 'in yeast'. The name of enzyme usually ends in – ase. Example: Cytochrome

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